Highlights

Steady-state compositional gas–liquid flow in production pipelines. Part I: a differentiable simulator with cross-verified solvers

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- Flux-form compositional pipe-flow model written once, differentiated and vectorised with JAX
- No-slip and mechanistic models solved by three cross-verified methods (march, FD, FV)
- Finite volumes close momentum and energy balances to machine precision across phase changes
- Compared with Aspen HYSYS on phase envelope, 200 pressure-drop lines and an offshore system
- Vertical pressure drops fall inside the range of four HYSYS correlations in 88

Steady-state compositional gas–liquid flow in production pipelines. Part I: a differentiable simulator with cross-verified solvers

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ARTICLE INFO

Keywords:
multiphase flow
compositional simulation
mechanistic model
automatic differentiation
finite volumes
pipelines

ABSTRACT

Steady-state compositional simulation of gas–liquid flow in wells, flowlines and risers couples a cubic equation of state and an isothermal flash to mixture momentum and energy balances closed by homogeneous or flow-pattern-dependent mechanistic relations. This article, the first of two, presents a simulator in which the physics is written once, in flux form, and executed on an array backend with automatic differentiation (JAX): the Jacobian of the conserved fluxes, the derivatives of the flash with respect to pressure and temperature and the sparse Newton Jacobians are exact, and the compiled program is vectorised over cases. A no-slip and a mechanistic model are solved by an adaptive Runge–Kutta march and by finite-difference and finite-volume nodal systems; the three methods agree to 0.01 bar and 0.1 K, the finite volumes close the momentum and energy balances to machine precision, and Newton converges in one to three iterations across phase and flow-pattern changes. The thermodynamic kernel reproduces an independent open-source Peng–Robinson implementation to a relative 1e-6. Compared with Aspen HYSYS, the phase envelope of an 11-component fluid agrees to 0.7 bar in the median; in 100 vertical lines the mechanistic model falls inside the range of four HYSYS correlations in 88

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Nomenclature

A	pipe cross-sectional area (m^2)
D	internal diameter (m)
F	wall-friction pressure gradient ($\mathrm{Pa\,m^{-1}}$)
g	gravitational acceleration ($\mathrm{m\,s^{-2}}$)
h_k	specific enthalpy of phase k ($\mathrm{J\,kg^{-1}}$)
H_L	liquid holdup (–)
$\mathbf{J}$	Jacobian $\partial\boldsymbol{\Phi}/\partial\mathbf{u}$
$\dot{m}$, $\dot{m}_k$	total and phase mass flow rates ($\mathrm{kg\,s^{-1}}$)
P	pressure (Pa; tables and figures in bar)
Q_w	heat exchanged with the surroundings per unit length ($\mathrm{W\,m^{-1}}$)
$\mathbf{S}$	source vector
T	temperature (K)
U	overall heat-transfer coefficient ($\mathrm{W\,m^{-2}\,K^{-1}}$)
$\mathbf{u}$	state vector (P, T)
v_k	velocity of phase k ($\mathrm{m\,s^{-1}}$)
w_G	vapour mass fraction from the flash (–)
x	axial coordinate (m)
β	molar vapour fraction (–)
λ_L	no-slip liquid volume fraction (–)
μ_k	dynamic viscosity of phase k (Pa s)
ρ_k	density of phase k ($\mathrm{kg\,m^{-3}}$)
σ	surface tension ($\mathrm{N\,m^{-1}}$)
$\boldsymbol{\Phi}$	conserved flux vector
θ	pipe inclination from the horizontal (rad)

1. Introduction

Flow assurance in offshore production depends on steady-state models of gas–liquid flow along wells, flowlines and risers that predict pressure, temperature, phase fractions and flow pattern from the fluid composition and the boundary conditions. Commercial simulators such as Aspen HYSYS (Aspen Technology, Inc.) offer a choice of empirical and mechanistic correlations for this purpose (Beggs and Brill, 1973; Aziz, Govier and Fogarasi, 1972; Mandhane, Gregory and Aziz, 1974; Zhang, Wang, Sarica and Brill, 2003), and the mechanistic, flow-pattern-based closures of Taitel and Dukler (1976); Taitel, Barnea and Dukler (1980), Xiao, Shoham and Brill (1990) and Gomez, Shoham, Schmidt, Chokshi and Northug (2000) remain the basis of most pipeline models. Compositional formulations couple them to a cubic equation of state (Peng and Robinson, 1976) and an isothermal flash (Michelsen, 1982a,b) so that phase behaviour, condensation and heat transfer are resolved along the pipe; the formulation used here, with a homogeneous no-slip model and a mechanistic model with slip, was developed in the first author's master's thesis (Carvalho, 2022).

Two difficulties recur in this class of simulators. The first is numerical: the derivatives that drive both a marching solution (the Jacobian of the conserved fluxes with respect to pressure and temperature, which contains the derivatives of the flash) and the Newton iterations of a nodal solution are usually approximated by finite differences, each of which costs additional flash calculations and introduces noise that interferes with convergence where phases appear or disappear. The second is cost: the flash dominates the run time, and the thousands of evaluations needed for optimisation, uncertainty quantification or surrogate training multiply it.

This article presents a reimplementation of the compositional model designed around automatic differentiation and vectorisation (Bradbury, Frostig, Hawkins, Johnson, Leary, Maclaurin, Necula, Paszke, VanderPlas, Wanderman-Milne and Zhang, 2018). The contributions are: (i) a flux-form statement of both flow models on a single interface, with exact Jacobians and flash derivatives obtained by forward-mode automatic differentiation; (ii) three interchangeable solvers (adaptive march, finite differences, finite volumes) that verify each other on the same equations, with global balance diagnostics, and a verification of the thermodynamic kernel against an independent open-source implementation; (iii) a comparison with Aspen HYSYS on the phase envelope and properties of an 11-component fluid, on 200 horizontal and vertical pressure-drop lines and on an offshore well–flowline–riser system; and (iv) a vectorised

march that generates hundreds of converged profiles per minute. The companion paper (Carvalho, Costa, Nunes and Souza, 2026) uses this simulator as the data generator and as the residual provider of neural and physics-informed surrogate models.

Section 2 states the model, Section 3 the numerical methods and the differentiable implementation, Section 4 the verification, Section 5 the comparison with HYSYS, Section 6 an application to a gas-condensate pipeline and Section 7 the computational performance.

2. Mathematical model

2.1. Conservation equations in flux form

A single pipe of length L is described by segments of constant diameter D, roughness ε, inclination θ, overall heat-transfer coefficient U and ambient temperature T_{amb}. The total mass flow rate $\dot{m}$ is constant; the vapour and liquid rates follow from the flash, $\dot{m}_G = w_G\dot{m}$ and $\dot{m}_L = (1 - w_G)\dot{m}$. Both flow models considered here are written as

$$\frac{\mathrm{d}\boldsymbol{\Phi}}{\mathrm{d}x} = \mathbf{S}, \qquad \boldsymbol{\Phi} = \begin{pmatrix} \dfrac{\dot{m}_G v_G + \dot{m}_L v_L}{A} + P \\ \dfrac{\dot{m}_G h_G + \dot{m}_L h_L}{\dot{m}} \end{pmatrix}, \qquad \mathbf{S} = \begin{pmatrix} -F - \rho_m g \sin\theta \\ \dfrac{Q_w}{\dot{m}} - g\sin\theta \end{pmatrix}, \tag{1}$$

where the first component is the mixture momentum balance (momentum flux plus pressure, with wall friction F and the hydrostatic term with the in situ mixture density $\rho_m = \rho_G(1 - H_L) + \rho_L H_L$) and the second is the mixture energy balance in terms of the mass-weighted specific enthalpy, with the potential-energy term and the heat exchanged with the surroundings,

$$Q_w = -\pi D\, U\, (T - T_{\mathrm{amb}}). \tag{2}$$

The kinetic-energy contribution to the energy balance is neglected; it is available as an option. The state vector is $\mathbf{u} = (P, T)$: for a given composition and $\dot{m}$, every quantity in Eqs. (1)–(2) is a local function of $\mathbf{u}$ through the flash and the closure relations. The two models differ only in how the phase velocities and the friction term are closed.

2.2. Thermodynamic model

Phase equilibrium and phase properties come from the Peng–Robinson equation of state (Peng and Robinson, 1976),

$$P = \frac{RT}{v - b} - \frac{a\,\alpha(T)}{v(v + b) + b(v - b)}, \tag{3}$$

with the classical mixing rules and zero binary interaction parameters (Appendix A). The isothermal flash solves the Rachford–Rice equation (Rachford and Rice, 1952) with successive substitution of the equilibrium ratios followed by Newton polishing (Michelsen, 1982a,b), preceded by a stability test. Phase densities follow from the compressibility roots, enthalpies from the ideal-gas heat-capacity polynomials, the enthalpies of formation and the residual enthalpy of the equation of state, liquid viscosity from the corresponding-states correlation of Allan and Teja (1991), vapour viscosity from Pedersen and Fredenslund (1987) and surface tension from the parachor method. The component data are listed in Appendix B. The no-slip liquid volume fraction follows directly from the flash,

$$\lambda_L = \frac{\dot{m}_L/\rho_L}{\dot{m}_L/\rho_L + \dot{m}_G/\rho_G}. \tag{4}$$

2.3. Closure relations

In the no-slip model the two phases share the velocity $v = \dot{m}/(\rho_m A)$ with $\rho_m = \rho_G(1 - \lambda_L) + \rho_L\lambda_L$, and the friction term uses the Churchill friction factor (Churchill, 1977) with volume-weighted mixture density and viscosity. In the mechanistic model the flow pattern is identified by the Taitel–Dukler criteria (Taitel and Dukler, 1976; Taitel et al., 1980), and the holdup and the friction term follow from the pattern-specific momentum balances of Xiao et al. (1990) for stratified and annular flow and of Gomez et al. (2000) for slug flow, with bubble flow treated by a drift-flux relation; the in situ holdup

$$H_L = \frac{S\lambda_L}{1 - \lambda_L + S\lambda_L} \tag{5}$$

is expressed through the slip ratio $S = v_G/v_L$. The pattern map and the transitions are evaluated at every state, so that the pattern may change along the pipe.

3. Numerical methods and differentiable implementation

3.1. Three solvers on one interface

Because both models expose the same pair $(\mathbf{\Phi}, \mathbf{S})$ as functions of $\mathbf{u}$, the solution methods are written once and apply to both. Three are used.

Marching With $\mathbf{J} = \partial\mathbf{\Phi}/\partial\mathbf{u}$, Eq. (1) becomes the initial-value problem

$$\mathbf{J}(\mathbf{u})\,\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}x} = \mathbf{S}(\mathbf{u}), \qquad \mathbf{u}(0) = (P_{\mathrm{in}}, T_{\mathrm{in}}), \tag{6}$$

integrated with the embedded Dormand–Prince 5(4) pair (Dormand and Prince, 1980) with adaptive step, from output point to output point, with the error measured on the state scaled by (1 bar, 1 K). Expanding $\mathrm{d}\mathbf{\Phi}/\mathrm{d}x$ term by term and evaluating the partial derivatives of the flash outputs by one-sided finite differences costs three flashes per evaluation; here $\mathbf{J}$ is exact (Section 3.2) and a single flash is needed per right-hand-side evaluation. The converged equilibrium ratios of the last accepted step are carried as an auxiliary variable and seed the next flash, which halves the substitution iterations without changing the result.

Finite differences With P and T as unknowns at n nodes per segment, the flux derivative is discretised with the second-order operator $\mathbf{M}$ (centred inside, one-sided at the ends), $(\mathbf{M\Phi})_i = \mathbf{S}_i$. For the mechanistic model the phase mass rates are eliminated algebraically ($\dot{m}_G = w_G\dot{m}$), which keeps two unknowns per node.

Finite volumes Integrating Eq. (1) over each cell gives

$$\mathbf{\Phi}_{i+1} - \mathbf{\Phi}_i = \frac{h}{2}\left(\mathbf{S}_i + \mathbf{S}_{i+1}\right), \tag{7}$$

which is conservative by construction and never differentiates the fluxes across the kinks introduced by the dew point or by a change of flow pattern. Break points between segments are duplicated with continuity of P and T. Both nodal systems are solved by Newton's method,

$$\mathbf{J}_N(\mathbf{u}^k)\,\delta\mathbf{u} = -\mathbf{R}(\mathbf{u}^k), \qquad \mathbf{u}^{k+1} = \mathbf{u}^k + \delta\mathbf{u}, \tag{8}$$

with the exact banded Jacobian assembled by forward-mode automatic differentiation with graph colouring (bandwidth two for the finite-difference operator, one for the finite volumes), starting from the no-slip march interpolated at the nodes. Every run reports the global balances $(\mathbf{\Phi}_{\mathrm{out}} - \mathbf{\Phi}_{\mathrm{in}}) - \int \mathbf{S}\,\mathrm{d}x$ per segment, evaluated on the solver's own grid, as a diagnostic.

3.2. Automatic differentiation and vectorisation

The physics is written once against an injected array backend; the same code runs in NumPy, for inspection and small cases, and in JAX (Bradbury et al., 2018), where it is compiled just in time and transformed. Three transformations matter here. First, forward-mode linearisation gives $\mathbf{J}$ in Eq. (6) and the coloured Newton Jacobian in Eq. (8) exactly, at the cost of two tangent evaluations instead of three extra flashes. Second, the flash is itself differentiable: the successive-substitution iterations are excluded from the derivative graph and the Newton polishing steps, which solve the equilibrium conditions to machine precision, carry the derivatives of the vapour fraction and of the phase compositions with respect to P and T by the implicit function theorem, so that the tangents are those of the converged solution and not of a truncated iteration. Functions with restricted domains (square roots, logarithms, powers) are guarded before evaluation so that the tangents stay finite, and the flash runs inside the compiled program, since an eager call would be retraced at every step. Third, the compiled march is vectorised over cases with the pipe and inlet parameters as arguments, so that one compilation serves any number of cases with the same fluid and grid and a whole batch is evaluated in a single call. This vectorised march is the data generator of the companion paper (Carvalho et al., 2026), where the same compiled right-hand side also provides the residual of the governing equations for physics-informed surrogates.

Table 1
Problems, fluids and references.

Problem	Fluid and geometry	Conditions	Reference
Thermodynamic states	11-component fluid (CO_2, N_2, H_2S, C_1–nC_8)	159 (P,T) states inside the phase envelope, 1–111 bar, 147–526 K	HYSYS, Peng–Robinson
Pressure-drop lines	CH_4/C_{7+} 70/30 mol; horizontal and vertical, L = 100 m, adiabatic, 40 °C	D = 4, 6, 8, 10 in; P_{in} = 20–160 bar; $\dot{m}$ = 2–10 kg/s (200 lines)	HYSYS: Beggs & Brill, Tulsa, HTFS, Gregory–Aziz–Mandhane (horizontal) or Aziz–Govier–Fogarasi (vertical)
Offshore system	9 components with C_{7+}	well 1500 m vertical, flowline 6000 m horizontal, riser 300 m vertical; 170 bar, 80 °C, 10 kg/s	HYSYS: Tulsa, Beggs & Brill
Gas-condensate pipeline	C_1–nC_5 rich gas	L = 8 km horizontal, D = 12 in; 70 bar, 15 kg/s; T_{in} = 60, 80 °C	none (application)

Table 2
Flash of the 11-component fluid at 159 states: this work against an independent open-source Peng–Robinson implementation.

Property	States	R^2	Max. relative deviation
vapour density	159	1.00000000	1.6e-06
liquid density	159	1.00000000	3.2e-07
molar vapour fraction	159	1.00000000	5.2e-06

Same flash computed by the open-source thermo library (independent Peng–Robinson implementation, exact Ω_a and Ω_b, zero binary interaction parameters) at the 159 states of the HYSYS comparison.

3.3. Reproducibility

Each case is a self-contained JSON document (fluid with component properties, pipe profile, inlet conditions, model and solver options) and each recorded run stores the profiles together with the case and the software provenance. Every figure and table of this article is produced by one script from the recorded runs and the reference data.

4. Verification

Table 1 lists the problems used in this and the next sections; the fluid compositions are in Table B.2.

4.1. Thermodynamic kernel

The flash of the 11-component fluid was computed at the 159 two-phase states of the HYSYS comparison (Section 5.1) by the present implementation and by the open-source thermo library (Bell and Contributors, 2024), an independent Peng–Robinson implementation, with the same critical constants, acentric factors and zero interaction parameters. The two agree to $R^2 = 1$ to eight decimals and to relative deviations below 2×10^{-6} in the vapour and liquid densities and in the vapour fraction (Table 2), so that the equation of state, the stability test and the flash are verified independently of the flow models.

4.2. Cross-verification of the solvers

Table 3 gives, for the offshore system and for one case of the gas-condensate pipeline of Section 6, the pressure and temperature at the check points obtained with each model, each solver and each backend, together with the Newton iterations and the global balance residuals. For a given model the three solvers agree to 0.01 bar and 0.1 K, and NumPy and JAX agree to the printed digits, so that the implementation of the equations is verified independently of the discretisation. The finite-volume balances close to machine precision on their own grid; those of the march and of the finite differences are of the order of the integration tolerance or of the discretisation error (tens to hundreds of pascal over 170 bar, a relative 10^{-6} to 10^{-5}). Starting from the no-slip march, the nodal systems of the no-slip model converge in a single Newton iteration and those of the mechanistic model in two or three, including the changes of flow pattern along the system and the onset of condensation.

Table 3
Two models, three solvers and two backends on the offshore system and on case 3 of the gas-condensate pipeline.

Problem	Model	Solver	Backend	P (bar)	T (°C)	Newton / steps	Mom. (Pa)	Energy (J/kg)
offshore	no-slip	march	JAX	119.98 / 114.39 / 105.16	65.9 / 42.0 / 39.1	158 st.	-2e+01	+2e-01
offshore	no-slip	FD	JAX	119.98 / 114.39 / 105.15	65.9 / 42.0 / 39.1	1 it.	-2e+02	+2e+00
offshore	no-slip	FV	JAX	119.98 / 114.39 / 105.16	65.9 / 42.0 / 39.1	1 it.	+1e-06	+6e-07
offshore	no-slip	FV	NumPy	119.98 / 114.39 / 105.16	65.9 / 42.0 / 39.1	1 it.	-9e-06	+6e-07
offshore	mechanistic	march	JAX	115.62 / 102.28 / 91.48	65.5 / 41.2 / 38.0	158 st.	-4e+01	+1e-01
offshore	mechanistic	march	NumPy	115.62 / 102.28 / 91.48	65.5 / 41.2 / 38.0	158 st.	-4e+01	-7e-01
offshore	mechanistic	FD	JAX	115.62 / 102.27 / 91.48	65.5 / 41.2 / 38.0	3 it.	-4e+02	+2e+00
offshore	mechanistic	FV	JAX	115.62 / 102.28 / 91.48	65.5 / 41.2 / 38.0	3 it.	-3e-08	-7e-11
condensate 3	no-slip	march	JAX	69.15	26.9	91 st.	+1e+00	+2e+01
condensate 3	no-slip	FD	JAX	69.15	26.9	2 it.	+5e+00	+8e+01
condensate 3	no-slip	FV	JAX	69.15	26.9	1 it.	-9e-06	-8e-07
condensate 3	mechanistic	march	JAX	69.11	26.9	91 st.	-1e+01	+2e+01
condensate 3	mechanistic	FD	JAX	69.11	26.9	2 it.	+3e+01	+8e+01
condensate 3	mechanistic	FV	JAX	69.11	26.9	1 it.	+5e-03	+9e-03

Solvers: adaptive march (RK45; number of accepted steps), finite differences (FD) and finite volumes (FV) with Newton iterations; 81 nodes (offshore) and 64 nodes (condensate case 3). P and T at 1500 / 7500 / 7800 m (offshore) or at the outlet (condensate). Mom. and Energy: global balance residuals $(\Phi_{\text{out}} - \Phi_{\text{in}}) - \int S \, \mathrm{d}x$ on the solver's own grid.

5. Comparison with Aspen HYSYS

The HYSYS results used here were computed by the first author with Aspen HYSYS (Aspen Technology, Inc.), Peng–Robinson property package and pipe segments with the correlations listed in Table 1, for the same fluids and conditions (Carvalho, 2022). The C_{7+} pseudo-component of the pressure-drop and offshore fluids was characterised in HYSYS and its properties are listed in Table B.1. The pressure drops of the 200 lines are tabulated values; the phase envelope, the property distributions and the offshore profiles are available as plots and were digitised. All of them are provided as supplementary material.

5.1. Phase envelope and fluid properties

Fig. 1 compares the phase envelope of the 11-component fluid computed by the present flash with the HYSYS envelope: the bubble line agrees with a median deviation of 0.66 bar and the dew line of 0.69 bar at the same temperature, with the largest deviations (up to 11 bar on the bubble line) near the critical point, where both lines are steep and the digitisation is least precise. The 159 states are the two-phase nodes of a 20×20 grid between 1 and 140 bar and between 100 and 550 K (Fig. 1). The HYSYS values of the phase properties at these states are available only as the markers of parity plots, without the state each marker belongs to, so the comparison is of distributions: Fig. 2 plots the quantiles of the present values against those of HYSYS, and Table 4 gives the ratios of the medians and of the 10th and 90th percentiles. Overlapping markers merge in the digitisation, which biases the digitised distribution towards rare values, so the present 159 values are merged in the same way before the quantiles are computed. The vapour and liquid densities, the liquid enthalpy and the surface tension follow the diagonal, with median ratios of 0.95–1.00, the densest vapour states being slightly above it; the molar vapour fraction agrees with a median ratio of 1.04 when the values are compared without merging, the ratio of the table (1.31) being an artefact of the merging in this panel, where it leaves 43 markers against the 72 of the figure; the vapour enthalpy is offset by up to 0.5 MJ kg^{-1} in the middle of its range, which reflects the ideal-gas heat-capacity and formation-enthalpy data of the two packages rather than the equation of state, since the residual part is verified in Section 4.1. The vapour viscosity is about 12 % below HYSYS over the whole range, the difference between the corresponding-states method of Pedersen and Fredenslund (1987) and the HYSYS method, and the liquid viscosity agrees below about 1.5 mPa s with a higher tail.

5.2. Pressure drop in horizontal and vertical lines

The 200 lines are 100 m long, adiabatic, horizontal or vertical, with four diameters, five inlet pressures and five mass flow rates (Table 1), and carry a binary mixture of methane and the C_{7+} pseudo-component (70/30 mol). HYSYS computed each line with four correlations: Beggs and Brill (1973), the Tulsa unified model (Zhang et al., 2003), the HTFS liquid-slip method, and Gregory–Aziz–Mandhane (Mandhane et al., 1974) in the horizontal or Aziz–Govier–Fogarasi (Aziz et al., 1972) in the vertical lines. Fig. 3 shows, case by case, the range spanned by the four correlations and the pressure drop of the two present models, and Table 6 summarises the deviations.

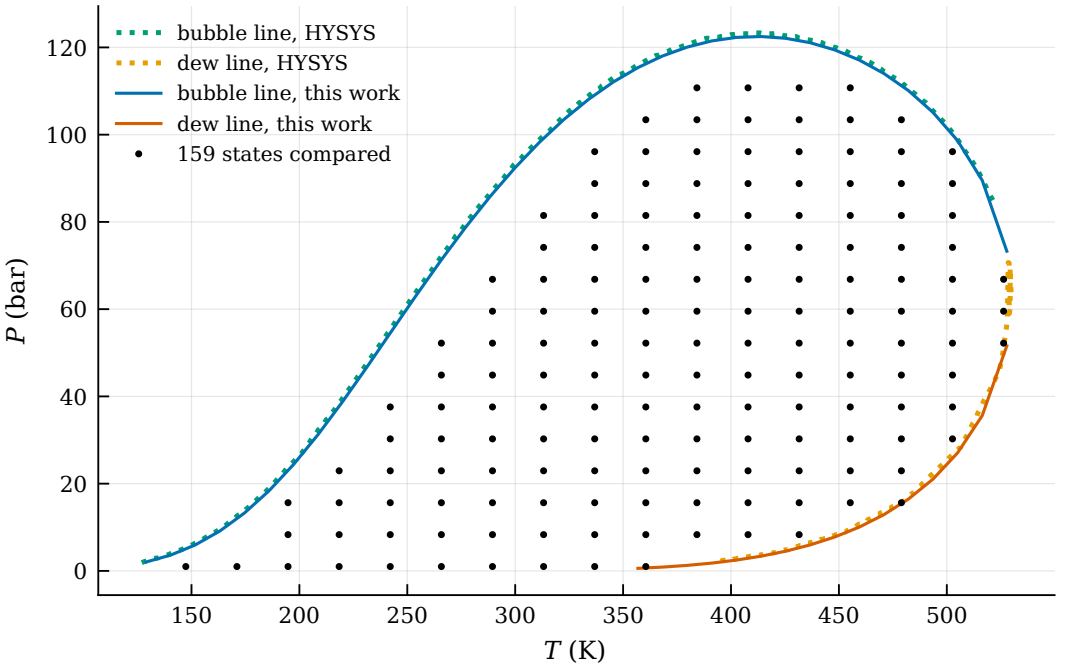


Figure 1: Phase envelope of the 11-component fluid: this work (solid) against HYSYS (dotted), and the 159 states of the property comparison.

Table 4
Phase envelope and property distributions: this work against HYSYS.

Property	Points: HYSYS / this work	Median ratio (this work / HYSYS)	10th / 90th percentile ratio
bubble pressure (bar)	–	median $\lvert\Delta\rvert = 0.66$, max 11.4	–
dew pressure (bar)	–	median $\lvert\Delta\rvert = 0.69$, max 4.8	–
ρ_V (kg m^{-3})	60 / 44	0.98	0.88 / 1.07
ρ_L (kg m^{-3})	91 / 52	0.97	0.97 / 1.00
h_V (MJ kg^{-1})	88 / 48	0.86	0.98 / 0.97
h_L (MJ kg^{-1})	85 / 53	0.95	1.01 / 0.98
β (–)	72 / 43	1.31	1.20 / 1.07
μ_V (μPa s)	86 / 39	0.88	0.89 / 0.93
μ_L (mPa s)	24 / 14	0.99	0.88 / 1.71
σ (mN m^{-1})	110 / 57	1.00	0.73 / 1.15

Envelope: deviation from the HYSYS bubble and dew lines at the same temperature. Properties: the HYSYS values are available only as unpaired markers of parity plots, so distributions are compared; overlapping markers merge in the digitisation and our 159 values are merged in the same way before computing quantiles.

In the vertical lines, where the hydrostatic term dominates and the four HYSYS correlations agree within 20 % of each other in the median, the mechanistic model falls inside their range in 88 % of the cases, with a median deviation of +1 % from their geometric mean, +2 % from Tulsa and −6 % from Beggs and Brill; the no-slip model, which cannot raise the holdup above the no-slip fraction, is 14 % lower, as expected, and drops out of the range at low pressure and large diameter, where slip matters most. In the horizontal lines the four HYSYS correlations themselves disagree by a factor of three in the median, and both present models lie below the whole range in 87–95 % of the cases: the mechanistic model is 52 % below Tulsa and 73 % below Beggs and Brill in the median. These lines are short, adiabatic and friction-dominated, with pressure drops from 10^{-4} to 1 bar, and their liquid is the heavy pseudo-component (560 g mol^{-1}), so they are sensitive both to the liquid viscosity and to the stratified-flow closure. The liquid viscosity given by the corresponding-states correlation is 0.7 mPa s to 3 mPa s at these conditions, decreasing with pressure as methane dissolves in the liquid; tripling it raises the computed pressure drops by about one fifth, so viscosity alone does not explain the gap.

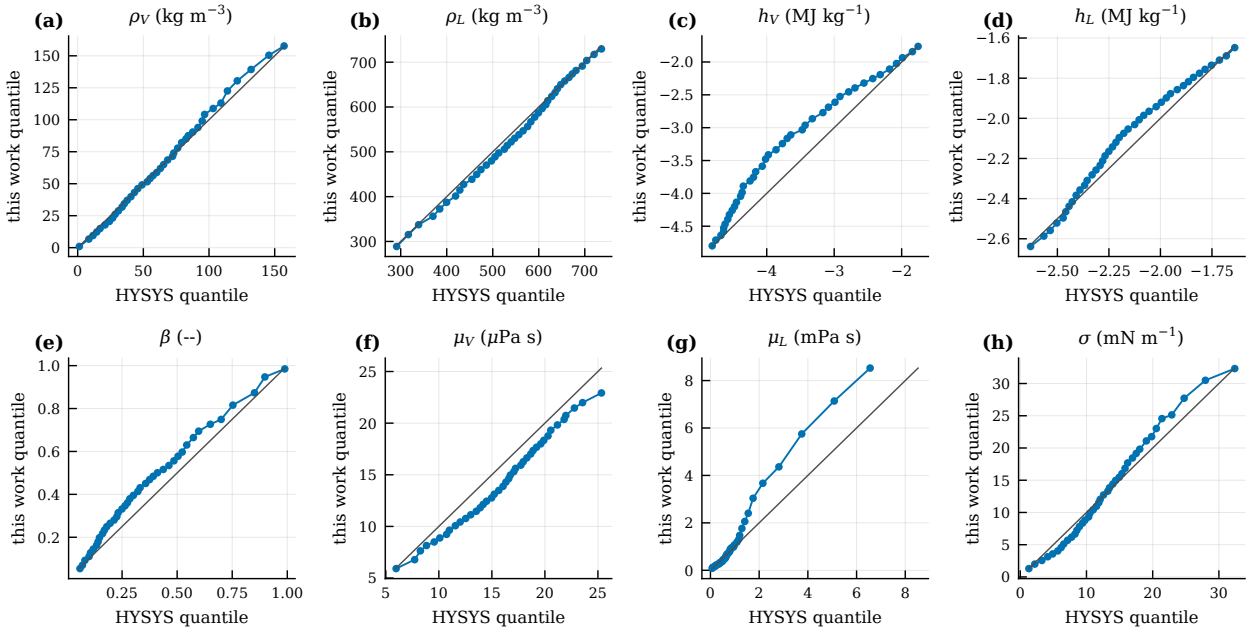


Figure 2: Quantile–quantile comparison of the phase properties at the 159 states with HYSYS: (a) vapour density, (b) liquid density, (c) vapour enthalpy, (d) liquid enthalpy, (e) molar vapour fraction, (f) vapour viscosity, (g) liquid viscosity and (h) surface tension; the line is the diagonal.

Table 5 isolates the other factors. The roughness of the HYSYS lines is not reported: raising it from 0.045 mm, close to the HYSYS default for steel pipe and assumed here, to 0.2 mm, the value of the original case definitions, increases the pressure drops by 14 % in the median and brings 11 % of the lines inside the range. The gap is concentrated in the lines that the mechanistic model classifies as stratified at the inlet (a third of them), where its pressure drop is 0.13 of the geometric mean of the correlations and 0.23 of Tulsa, against 0.35 and 0.58 in the lines in intermittent flow; it also widens with the diameter. In the stratified lines both phases move at 0.2 m s^{-1} to 0.5 m s^{-1}, the liquid fills about 70 % of the section and the pressure drop over 100 m is 0.1 to 10 mbar, so the result rests entirely on the wall and interfacial shear of the Taitel–Dukler balance; the correlations, with their own holdup and pattern maps, imply frictional gradients 3.7 times larger in the median. That factor is close to the ratio of the Darcy to the Fanning friction factor: using a Darcy factor in the Fanning-form wall shear places 44 % of the lines inside the HYSYS range and 34 % above it. Agreement of a mechanistic model with these correlations in friction-dominated horizontal flow is therefore weak evidence on its own, since a convention error of exactly this size would produce it; the present implementation uses the Fanning factor throughout, consistently with the vertical lines and the offshore system, where it agrees with HYSYS. With the information available, the horizontal gap reflects the difference between the mechanistic closures and the empirical correlations for a slow, heavy liquid layer, a difference that only measurements, or the HYSYS holdups and flow patterns for the same states, could arbitrate.

5.3. Offshore production system

The offshore system consists of a vertical well of 1500 m and 127 mm, a horizontal flowline of 6000 m and a vertical riser of 300 m, both of 152.4 mm, with overall heat-transfer coefficients of 7 and 5 W m^{-2} K^{-1} and ambient temperatures of 40 and 12 °C; a 9-component fluid with the C_{7+} pseudo-component enters the well at 170 bar, 80 °C and 10 kg s^{-1}. HYSYS computed the system with the Tulsa and Beggs and Brill correlations. Fig. 4 and Table 7 compare the profiles. The two HYSYS correlations differ by up to 10 bar in pressure, and the present models fall on either side: the no-slip model follows Tulsa in the well and in the first half of the flowline (120.0 bar against 118.7 bar at the wellhead, 117.1 bar against 117.8 bar at mid-flowline), while the mechanistic model loses more pressure in the well (115.6 bar at the wellhead, between Tulsa and Beggs and Brill) and then follows the Beggs and Brill profile within about 1 bar along the flowline and at the riser base (102.3 bar against 102.6 bar). The temperatures agree within 2 K everywhere, the present ones being 0.1 to 1.7 K warmer. The no-slip holdup, the flash fraction λ_L, is below both

Table 5
Horizontal lines: sensitivity of the pressure drop to the roughness and to the friction-factor convention, and the gap by flow pattern.

Model	Variant	Lines	Inside (%)	Above (%)	ΔP / HYSYS geom. mean	ΔP / Tulsa	ΔP / base
mechanistic	base: $\varepsilon = 0.045$ mm, Fanning wall friction	100	5	0	0.27	0.48	–
mechanistic	$\varepsilon = 0.2$ mm	100	11	0	0.32	0.56	1.14
mechanistic	wall friction ×4 (Darcy factor in Fanning-form shear)	100	44	34	1.44	2.17	5.20
mechanistic	base, lines in stratified flow	33	0	0	0.13	0.23	–
mechanistic	base, lines in intermittent flow	67	7	0	0.35	0.58	–
no-slip	base: $\varepsilon = 0.045$ mm	100	13	0	0.34	0.54	–
no-slip	$\varepsilon = 0.2$ mm	100	18	0	0.38	0.62	1.14

Medians over the 100 horizontal lines (or over the lines of each flow pattern, as classified by the mechanistic model at the inlet). The roughness of the HYSYS lines is not reported; 0.045 mm is close to the HYSYS default for steel pipe and 0.2 mm is the value of the original case definitions. The factor 4 on the wall friction is the ratio of the Darcy to the Fanning friction factor.

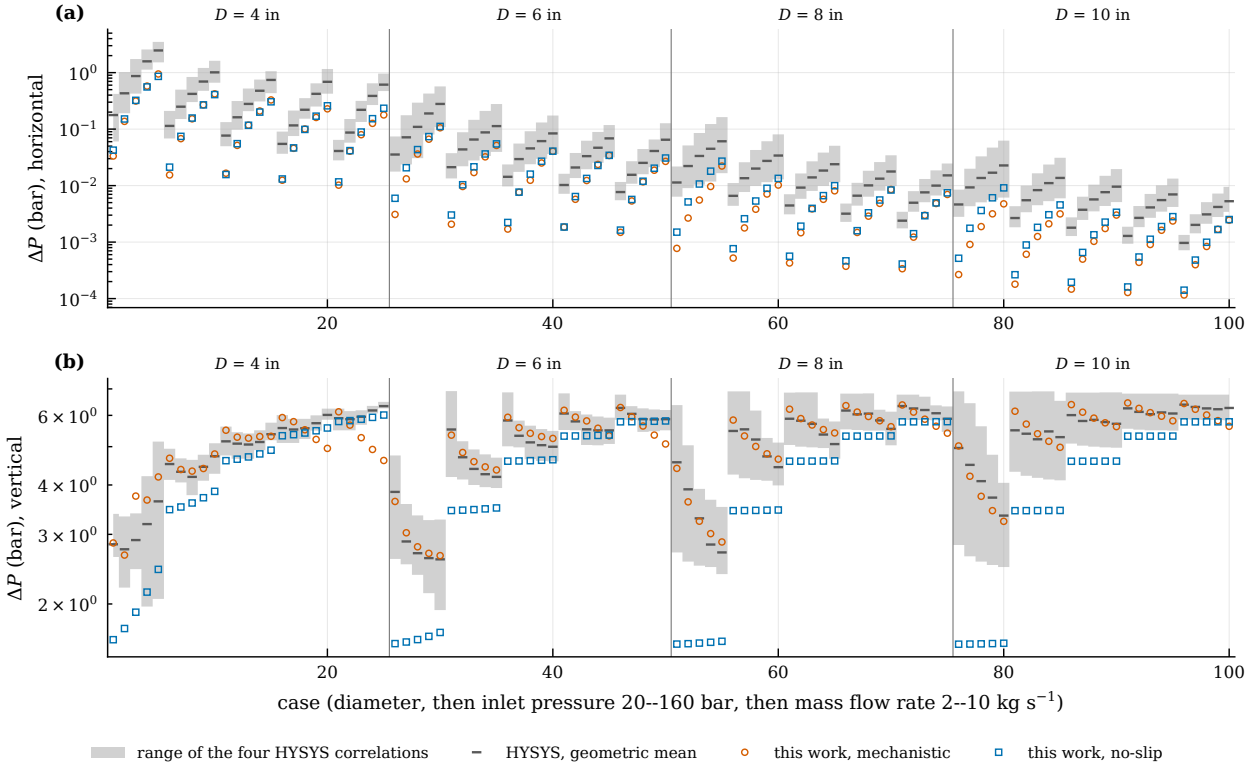


Figure 3: Pressure drop in the (a) horizontal and (b) vertical lines, case by case: range of the four HYSYS correlations (grey band) and their geometric mean (dash), and the mechanistic (circles) and no-slip (squares) models of this work.

HYSYS holdups, as it must be when the gas moves faster than the liquid; the mechanistic holdup lies between the two correlations in the well, below them in the flowline (0.34 against 0.38 and 0.45) and above them at the riser base. The mechanistic model predicts annular flow along the whole system, whereas the Beggs and Brill correlation in HYSYS reports intermittent flow throughout, a difference of classification that the pressure profiles absorb.

6. Application: condensation in a gas pipeline

An 8 km horizontal pipeline of 12 in carries a rich gas of methane to *n*-pentane (Table B.2) at 70 bar and 15 kg s^{-1}, with the ambient at 25 °C. Six cases combine two inlet temperatures (60 and 80 °C) with three overall heat-transfer coefficients (3.4, 10.3 and 20.5 W m^{-2} K^{-1}, from an insulated to a bare line), and the question is whether and where condensate forms. Fig. 5 and Table 8 show the results of the no-slip model. With the lowest coefficient the gas reaches the outlet at 43 and 54 °C and stays single-phase. With the two higher coefficients it reaches the dew point between

Table 6
Pressure drop in the 200 lines: this work against the four HYSYS correlations.

Orientation	Model	Inside HYSYS range (%)	Below (%)	vs. geom. mean (%)	vs. B&B (%)	vs. Tulsa (%)	vs. HTFS (%)	vs. Gregory / Aziz (%)
horizontal	no-slip	13	87	-66	-69	-46	-67	-83
horizontal	mechanistic	5	95	-73	-73	-52	-71	-86
vertical	no-slip	19	81	-14	-20	-11	-8	-19
vertical	mechanistic	88	11	+1	-6	+2	+9	-6

Median deviation $\Delta P_{\text{this work}}/\Delta P_{\text{HYSYS}} - 1$ over 100 lines per orientation. HYSYS correlations: Beggs & Brill / Tulsa / HTFS / Gregory–Aziz–Mandhane (horizontal); Beggs & Brill / Tulsa / HTFS / Aziz–Govier–Fogarasi (vertical). The spread of the four correlations themselves (max/min) has a median of 3.0 in the horizontal and 1.2 in the vertical lines.

Table 7
Offshore system: pressure, temperature and holdup at the wellhead, mid-flowline and riser base, this work against HYSYS.

Position	Quantity	This work, no-slip	This work, mechanistic	HYSYS, Tulsa	HYSYS, Beggs & Brill
wellhead (1500 m)	P (bar)	120.0	115.6	118.7	112.5
wellhead (1500 m)	T (°C)	65.9	65.5	65.0	64.5
wellhead (1500 m)	H_L (–)	0.324	0.334	0.348	0.398
mid-flowline (4500 m)	P (bar)	117.1	108.7	117.8	107.4
mid-flowline (4500 m)	T (°C)	52.2	51.6	51.2	50.6
mid-flowline (4500 m)	H_L (–)	0.344	0.335	0.378	0.447
riser base (7500 m)	P (bar)	114.4	102.3	–	102.6
riser base (7500 m)	T (°C)	42.0	41.2	41.1	40.3
riser base (7500 m)	H_L (–)	0.360	0.472	0.400	0.445

HYSYS values read from the digitised profiles; digitisation uncertainty about 0.3 bar, 0.2 K and 0.005 in holdup. The dotted Tulsa curve is smoothed by a 9-point running median in temperature and holdup; its pressure is too sparsely digitised beyond 4.5 km to be read at the riser base.

Table 8
Gas-condensate pipeline: outlet conditions and onset of condensation.

Case	T_{in} (°C)	U (W m^{-2} K^{-1})	T_{out} (°C)	P_{out} (bar)	Onset of condensation (m)	T at onset (°C)	β_{out}
1	60	3.4	43.4	69.04	–	–	1.000
2	60	10.3	31.5	69.11	5800	34.2	0.985
3	60	20.5	26.9	69.15	3000	34.0	0.961
4	80	3.4	53.6	68.95	–	–	1.000
5	80	10.3	33.7	69.05	7600	34.3	0.997
6	80	20.5	27.5	69.12	3900	34.1	0.965

No-slip model, adaptive march; 70 bar and 15 kg/s at the inlet, ambient at 25 °C.

3.0 and 7.6 km, always at 34.0 °C to 34.3 °C, the dew temperature of the fluid at the local pressure, and leaves with 0.3 to 3.9 mol% of liquid; the higher inlet temperature only delays the onset. The march resolves the onset without special treatment, since the flash and its derivatives are continuous across the dew point, and the finite-volume solver reproduces the same profiles (Table 3, case 3).

7. Computational performance

Table 9 lists wall times on the offshore system. The first run of each model structure includes the just-in-time compilation, 13 s to 38 s, which is paid once and reused for any other inlet condition, pipe parameter or flow rate. Once compiled, the nodal systems with 81 nodes take 0.5 s to 0.7 s, one to three Newton iterations with the exact coloured Jacobian, and the adaptive march takes 1.2 s for the no-slip and 4.1 s for the mechanistic model, with 158 accepted steps and 157 output points. The NumPy backend, which evaluates the same equations without compilation and with finite-difference derivatives, takes 15 s for the finite-volume system, 25 times the compiled JAX run, and 73 s for the mechanistic march. For many cases of one segment the compilation dominates: a march of the well takes 46 s the first time and 0.25 s once compiled, and the vectorised march evaluates 300 cases in 112 s including compilation and in 54 s once compiled, 0.18 s per case. On a CPU, vectorisation thus gains about 30 % per case over compiled calls made

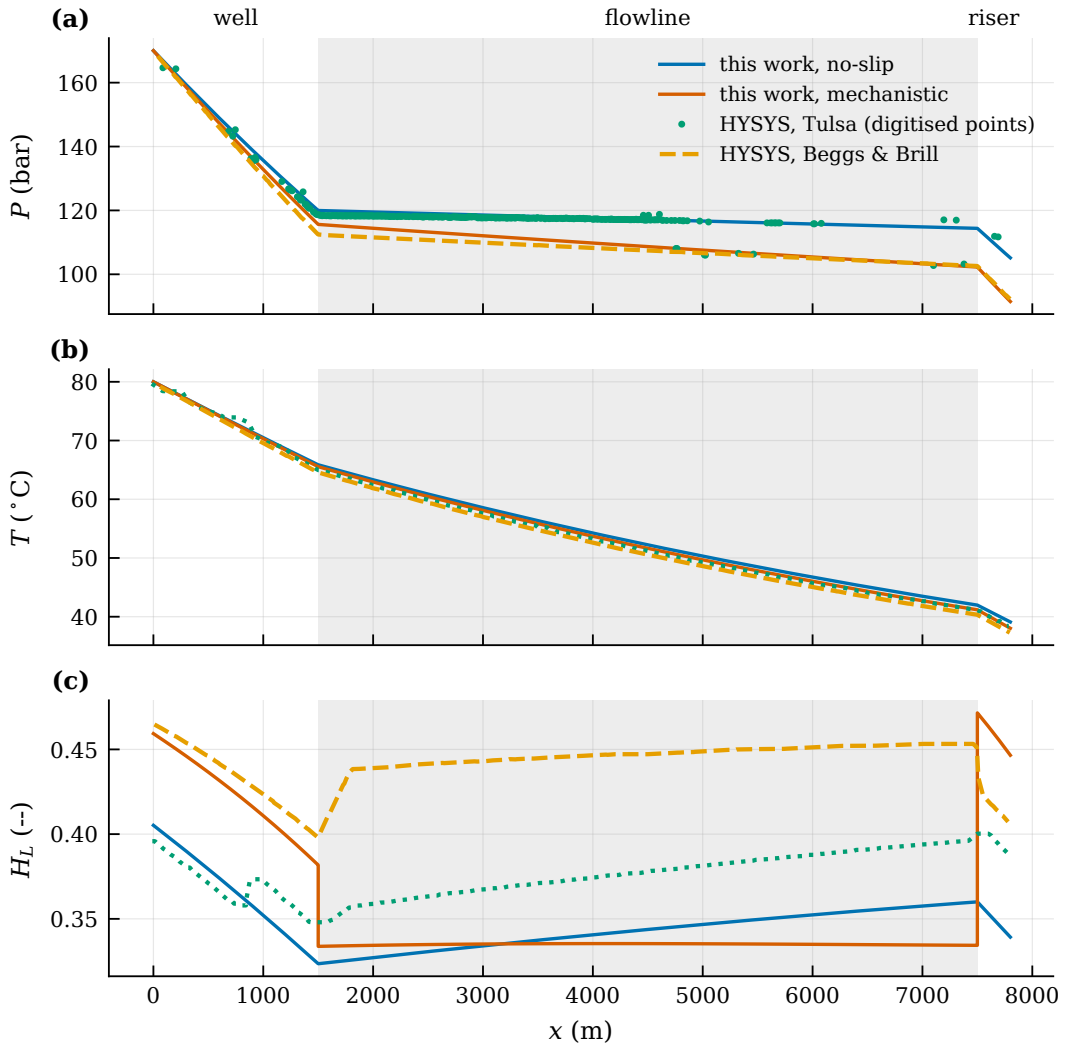


Figure 4: Profiles of (a) pressure, (b) temperature and (c) liquid holdup along the well–flowline–riser system (shaded: flowline): the no-slip and mechanistic models of this work (solid) against HYSYS with the Tulsa (dotted; pressure as digitised points) and Beggs and Brill (dashed) correlations.

one by one; its main benefit is that a single compilation serves the whole batch, which is how the companion paper generates its training sets in about a minute per segment.

8. Conclusions

Writing the compositional two-phase pipe-flow model in flux form on an array backend with automatic differentiation gave exact Jacobians and flash derivatives, three solvers that verify one another to 0.01 bar on the same equations, global balances that close to machine precision in the finite-volume scheme, Newton convergence in one to three iterations across phase and pattern changes, and a vectorised march that generates hundreds of converged profiles per minute. The thermodynamic kernel reproduces an independent open-source Peng–Robinson implementation to a relative 10^{-6} and the HYSYS phase envelope to 0.7 bar in the median. Against the HYSYS pipe correlations, the mechanistic model falls inside their range in 88 % of 100 vertical lines and follows the Beggs and Brill profile of an offshore system within about 1 bar along the flowline, with temperatures within 2 K; in short horizontal lines carrying a heavy pseudo-component both models lie below the range of the HYSYS correlations, which themselves span a factor of three, by a median factor of 3.7 concentrated in stratified flow; a rougher wall or a more viscous liquid closes only a

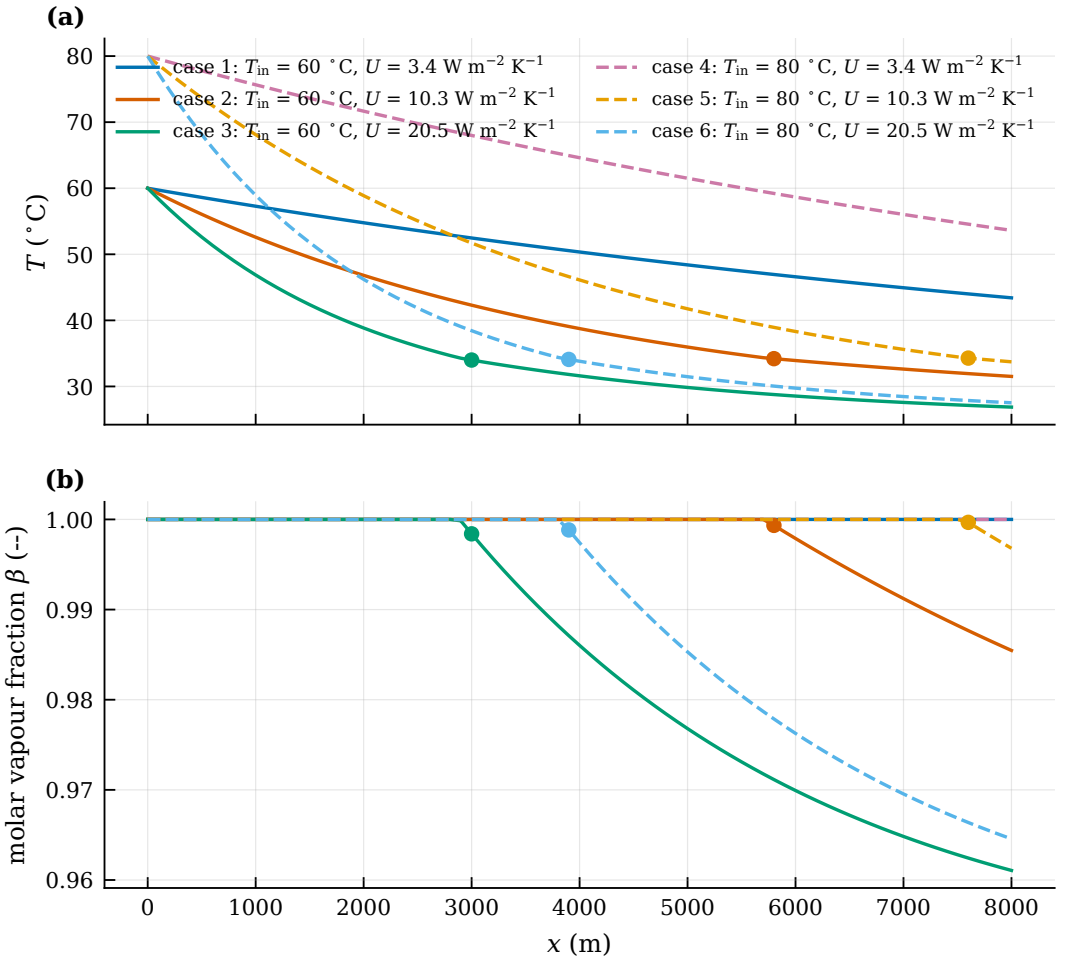


Figure 5: Gas-condensate pipeline: (a) temperature and (b) molar vapour fraction along the line for the six cases (solid: inlet at 60 °C; dashed: 80 °C); circles mark the onset of condensation.

fraction of it, and since the factor is that between the Darcy and Fanning friction factors, agreement of a mechanistic model with these correlations in friction-dominated horizontal flow is weak evidence of correct wall friction. The finite-volume scheme on the flux interface is the recommended nodal method, since it is conservative, narrow-banded and insensitive to the kinks of the dew point and of the pattern transitions, while the march remains the tool of choice for generating data. The companion paper (Carvalho et al., 2026) builds on both: the vectorised march generates the training sets of neural surrogates of the production system, and the compiled right-hand side supplies the residual of the governing equations to physics-informed surrogates that return whole profiles with the axial position as an input.

Acknowledgements

The authors thank the Programa de Pós-Graduação em Engenharia Química of the Universidade do Estado do Rio de Janeiro.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 9
Wall times on the offshore system.

Model	Solver	Newton / steps	JAX, first run (s)	JAX, compiled (s)	NumPy (s)
no-slip	march (RK45)	158 st.	16.0	1.26	–
no-slip	finite differences	1 it.	28.2	0.58	–
no-slip	finite volumes	1 it.	12.7	0.56	14.7
mechanistic	march (RK45)	158 st.	38.4	4.13	73.3
mechanistic	finite differences	3 it.	15.7	0.65	–
mechanistic	finite volumes	3 it.	28.3	0.70	–
no-slip	march, well segment, one case	–	46.1	0.25	–
no-slip	vectorised march, well segment, 300 cases	–	112.3	54.2 (0.18 per case)	–

Offshore system (81 nodes for the nodal solvers); wall times on an Apple silicon laptop, CPU only. The first JAX run includes just-in-time compilation, which is paid once per model structure (fluid, options and grid) and reused for any other inlet condition, pipe parameter or flow rate.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for code development, for the digitisation of plotted reference data and for language editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator, the case definitions, the recorded runs, the script that produces every figure and table of this article, and the HYSYS reference values (the 200 pressure drops and the digitised offshore profiles, as CSV files) are available in a public repository and as supplementary material.

A. Equation-of-state parameters

The attractive and covolume parameters of Eq. (3) are

$$a = \Omega_a \frac{R^2 T_c^2}{P_c}, \qquad b = \Omega_b \frac{R T_c}{P_c},$$
$$\alpha(T) = \left[1 + \kappa\left(1 - \sqrt{T/T_c}\right)\right]^2, \qquad \kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2, \tag{A.1}$$

with $\Omega_a = 0.45724$ and $\Omega_b = 0.0778$, and the van der Waals mixing rules with zero binary interaction parameters. The comparison with the open-source implementation in Section 4.1 uses the exact values of Ω_a and Ω_b in both codes; the rounded values change the properties by about 10^{-5}.

B. Component data and fluid compositions

Table B.1 lists the pure-component data and Table B.2 the compositions of the four fluids.

Table B.1
Component data.

Component	M (g/mol)	T_c (K)	P_c (bar)	ω	T_b (K)	Δh_f (kJ/mol)	A	$10^3 B$	$10^6 C$	$10^{-5} D$
CO_2	44.0	304.1	73.70	0.2389	194.5	-393.79	5.457	1.045	0.000	-1.157
N_2	28.0	126.2	33.90	0.0400	77.3	0.00	3.280	0.593	0.000	0.040
H_2S	34.1	373.5	89.60	0.0940	212.8	-20.63	3.931	1.490	0.000	-0.232
C_1	16.0	190.6	46.00	0.0120	111.4	-74.90	1.702	9.081	-2.164	0.000
C_2	30.1	305.4	48.80	0.0986	184.6	-84.74	1.131	19.225	-5.561	0.000
C_3	44.1	369.8	42.50	0.1520	231.1	-103.89	1.213	28.785	-8.824	0.000
n-C_4	58.1	425.1	38.00	0.2000	272.7	-126.19	1.935	36.915	-11.402	0.000
n-C_5	72.2	469.7	33.70	0.2520	309.2	-146.49	2.464	45.351	-14.111	0.000
n-C_6	86.2	507.9	30.30	0.3007	341.9	-167.29	3.025	53.722	-16.791	0.000
n-C_7	100.2	540.2	27.40	0.3500	371.6	-187.89	3.570	62.127	-19.486	0.000
n-C_8	114.2	568.7	24.90	0.4000	398.8	-208.59	4.108	70.567	-22.208	0.000
C_{7+}	560.0	991.4	8.38	1.3200	841.9	-572.88	21.400	340.000	-108.600	0.000

Ideal-gas heat capacity $c_p^{\mathrm{ig}}/R = A + BT + CT^2 + DT^{-2}$ (T in K). The C_{7+} pseudo-component was characterised in HYSYS (molar mass, density 950 kg/m^3, T_b, T_c, P_c, ω and enthalpy of formation); its ideal-gas heat capacity is extrapolated from the normal alkanes.

Table B.2
Fluid compositions.

Component	Thermodynamic states	Pressure-drop lines	Offshore system	Gas-condensate pipeline
CO_2	0.9	–	0.7	–
N_2	0.7	–	1.1	–
H_2S	0.1	–	–	–
C_1	38.4	70.0	69.6	71.4
C_2	1.6	–	10.3	15.2
C_3	2.6	–	4.9	7.8
n-C_4	3.0	–	3.2	3.9
n-C_5	2.6	–	2.2	1.7
n-C_6	2.4	–	0.7	–
n-C_7	3.4	–	–	–
n-C_8	44.3	–	–	–
C_{7+}	–	30.0	7.3	–

Mole percent.

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